

Graphs of inverse functions (0.1.4)

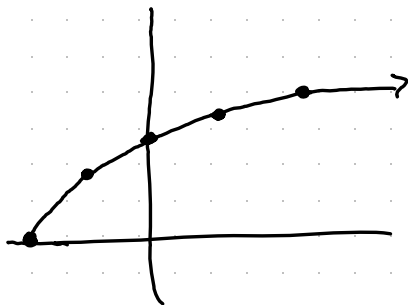
Definition: Given a function f with domain D and range R , the inverse of f (if it exists) is the function f^{-1} with domain R and range D such that $f(x) = y$ if and only if $f^{-1}(y) = x$.

Equivalently, $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Example: $f(x) = \sqrt{x+2}$

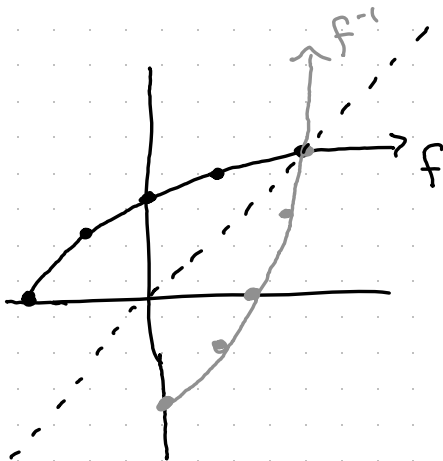
domain is $[-2, \infty)$, range $[0, \infty)$

x	$f(x)$
-2	0
-1	1
0	$\sqrt{2}$
1	$\sqrt{3}$
2	2



f^{-1} has range $[-2, \infty)$
and domain $[0, \infty)$

x	$f(x)$
-2	0
-1	1
0	$\sqrt{2}$
1	$\sqrt{3}$
2	2



x	$f^{-1}(x)$
0	-2
1	-1
$\sqrt{2}$	0
$\sqrt{3}$	1
2	2

The graph of f^{-1} is the graph of f reflected across the line $y=x$.

$$\sqrt{x+2} = y \iff x+2 = y^2 \iff x = y^2 - 2$$

$f^{-1}(x) = x^2 - 2$ on the domain $[0, \infty)$.